



Incident Escalation Practice Report

1. Introduction

Incident escalation is an essential process in Security Operations Center (SOC) operations that ensures security incidents are handled by the appropriate level of expertise based on severity and impact. Since not all security alerts represent the same level of risk, escalation workflows help prioritize incidents, improve response efficiency, and minimize organizational damage.

In this task, a simulated security incident involving unauthorized access was analyzed and escalated using **TheHive case management system**. A structured escalation workflow was followed, including incident validation, severity assessment, situation reporting (SITREP), and automated workflow assignment. The incident was mapped to attacker techniques defined by the **MITRE ATT&CK framework** to support standardized threat classification.

This exercise demonstrates how SOC teams coordinate incident handling across different response tiers and communicate effectively during security events.

2. Objective

The objective of this task is to simulate incident escalation procedures within a SOC environment and understand structured communication and response workflows.

Key Objectives

- Create incident cases in case management systems.
 - Evaluate incident severity and impact.
 - Escalate incidents to appropriate SOC tiers.
 - Draft Situation Reports (SITREP).
 - Implement automated escalation workflows.
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3. Tools Used

- TheHive - Incident case management and escalation
 - Google Docs - Incident reporting and SITREP documentation
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4. SOC Escalation Workflow Overview

Incident escalation ensures that alerts requiring deeper investigation are transferred from lower SOC tiers to higher levels of expertise.

SOC Tier Structure

Tier Level	Role	Responsibilities
Tier 1 (T1)	SOC Analyst	Alert monitoring and initial triage
Tier 2 (T2)	SOC Investigator	Deep analysis and containment
Tier 3 (T3)	Threat Response Team	Advanced forensics and remediation
Management	SOC Manager / CISO	Strategic decision-making

Escalation Criteria

- Incident severity
 - Impact on organizational systems
 - Scope of compromise
 - Confidence of malicious activity
 - Technical complexity
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5. Methodology / Implementation

5.1 Incident Identification

A simulated high-priority security alert representing unauthorized access on a critical server was generated. The alert indicated suspicious authentication activity from an unknown IP address.

Incident Details

- Incident Type: Unauthorized system access
- Detection Time: 2025-08-18 13:00
- Source IP: 192.168.1.200
- Affected Asset: Server-Y
- Threat Classification: Valid account misuse

The activity suggested potential credential compromise and required escalation.

5.2 Case Creation in TheHive

A new incident case was created in TheHive for tracking investigation activities.



Incident Case Details

Field	Value
Case Title	Unauthorized Access on Server-Y
Severity	High
Status	Open
Assigned Team	SOC Tier-2
Detection Source	Security monitoring system
Threat Technique	T1078 – Valid Accounts

The case was assigned high severity due to possible system compromise.

5.3 Escalation Simulation

A high-severity security incident involving unauthorized access to Server-Y was detected at 13:00 from IP address 192.168.1.200. Initial investigation confirmed suspicious authentication activity inconsistent with normal user behavior. The incident indicates possible credential compromise and unauthorized system access mapped to MITRE ATT&CK technique T1078. Immediate containment actions included isolating the affected server and collecting authentication logs. Due to the potential impact on critical systems and confirmed malicious indicators, the incident has been escalated to Tier-2 for detailed investigation, forensic analysis, and remediation.

5.4 Situation Report (SITREP)

SITREP Title: Unauthorized Access on Server-Y

Summary:

Unauthorized access activity was detected on Server-Y at 2025-08-18 13:00 from IP address 192.168.1.200. The activity involved abnormal login behavior mapped to MITRE technique T1078.

Actions Taken:

- Investigated authentication logs.
- Isolated affected server.
- Escalated incident to Tier-2 investigation team.

Next Steps:

- Perform forensic analysis.
- Reset compromised credentials.
- Monitor affected systems.



5.5 Workflow Automation

A simple automated workflow was designed to assign high-priority alerts to Tier-2 analysts automatically.

Automation Process

- Detect high-severity alert.
- Generate incident case.
- Assign case to Tier-2 team.
- Notify security personnel.

Automation reduces response time and ensures consistent incident handling.

6. Results and Findings

The incident escalation process produced the following results:

- Successful identification of unauthorized access.
- Proper classification of incident severity.
- Efficient case creation and tracking.
- Structured escalation to higher SOC tier.
- Improved incident response coordination.

The workflow ensured that the incident received appropriate investigation and response.

7. Security Impact

Effective incident escalation provides several operational benefits:

- Faster response to critical threats.
 - Clear communication across security teams.
 - Reduced risk of system compromise.
 - Improved incident management efficiency.
 - Better accountability and documentation.
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8. Conclusion

This task demonstrated incident escalation procedures within a SOC environment using structured workflows and case management practices. The simulation showed how incidents are identified, classified, escalated, and communicated across SOC tiers. Effective escalation ensures timely response, minimizes security risks, and improves overall organizational resilience against cyber threats.